# Xinyu (Catherine) Tian

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## EDUCATION

*Jul 2026 – Jan 2027*

**University of California San Diego** La Jolla, CA
*Visiting Student · Swartz Center for Computational Neuroscience, Jung Lab*

*Sep 2023 – Jun 2027*

**Tsinghua University** Beijing, China
*B.S. in Biology, Zhili College · GPA 3.9 / 4.0 · rank 3 / 35*

- Advisors: Prof. Xiaorong Gao, Prof. Bo Hong

## PUBLICATIONS

*2026*

**GazeAgent: VLM-Guided Implicit Gaze Calibration from Egocentric Multimodal Behavior for AI Glasses**
*Yuhong Zhang, Khanh An Nguyen, **Xinyu Tian**, Yuchen Xu, KungYi Chang, Abraham Padua, Ying Wu, Gert Cauwenberghs, Tzyy-Ping Jung · Under review at ACM CHI*

## RESEARCH

*Egocentric Gaze and Vision–Language Models*

*Apr – Sep 2026*

**Deriving Pseudo-Calibration Targets from Precise Visual Attention**
*UC San Diego, Jung Lab (remote Apr – Jun) · Part of GazeAgent (CHI submission, third author)*

- Benchmarked nine VLMs on validating precise-visual-attention events and localizing their targets from egocentric video, gaze and IMU streams
- Fine-tuned Qwen3.5-35B-A3B with LoRA on human-consensus labels, raising validation accuracy from 75.0% to 89.5% (Claude Opus 4.6: 81.4%)
- Designed the recall-oriented multimodal detector and the annotation criteria; five annotators converged to a median of 1.93° on targets of median width 13°

*Apr – Jun 2026*

**Is Egocentric Gaze Autoregressive? Toward a Gaze Foundation Model**
*Independent project (remote)*

- Built a gaze-tokenization pipeline on Microsoft HoloAssist: projected HoloLens gaze into RGB frames via depth (758k samples, 143 sessions), with five gaze representations and eight tokenizers feeding transformer sequence models
- Found next-fixation prediction dominated by temporal continuity: the best model beat a persistence baseline by only 3.2%, and instruction text added no gain, suggesting gaze needs external task context rather than autoregression
- This finding informed the context-driven design of GazeAgent above

*Jul 2026 – present*

**Causal Alignment of Human Gaze and Vision–Language Model Attention**
*UC San Diego* · GROUP LEADER

- Tested whether human fixation causally drives model attention in LLaVA and Qwen2-VL, patching activations at gaze-peak tokens against ground-truth fixations from the VQA-MHUG corpus
- Ablating gaze-peak tokens produced a significantly larger effect than matched random ablation (preliminary; $p < 0.01$, $N = 120$)
- Implemented in TransformerLens, following the ARENA 3.0 mechanistic-interpretability methodology

*Brain–Computer Interfaces for Real-Time Control*

*Feb 2025 – Jan 2026*

### Low-Fatigue Stimulus Design for SSVEP Spellers

*Tsinghua University, Gao Lab* · GROUP LEADER

- Replaced high-depth brightness flicker with low-depth and text-based stimuli, targeting the flicker sensation that limits how long a speller can be used rather than its peak accuracy
- Encoded three levels of Chinese linguistic structure — zi (character), ci (word) and chengyu (phrase) — to improve decoding robustness across units of different length
- Benchmarked FBCCA, TDCA and TRCA on accuracy and runtime across EEG from ten subjects; the low-depth study was conducted independently

*Jun – Sep 2025*

### Real-Time Wheelchair Navigation via a Text-Based SSVEP Interface

*Tsinghua University, Gao Lab* · GROUP MEMBER

- Built a closed-loop interface issuing seven directional commands — forward, back, left, right, two rotations and stop — at 0.5 s decision intervals using calibrated TDCA
- Presented text-coded stimuli through an AR headset, trading peak decoding accuracy for stimuli a user can tolerate across a full session
- Validated and debugged the deployed system for the World Robot Contest

*Computer Vision and Signal Processing for Clinical Assessment*

*Jan – Mar 2026*

### Automatic Detection of SPES Artifacts in ECoG

*King's College London* · GROUP LEADER

- Built an EEGLAB plugin that flags single-pulse electrical stimulation artifacts for seizure-onset-zone identification, replacing exhaustive visual inspection with a human-in-the-loop review step
- Across the five clinical recordings available, tuned detection caught every artifact with under 5% false positives and cut review time by roughly 80% (approximate figures on a small sample)
- Deliberately rule-based rather than learned, so a clinician can predict and audit its behaviour

*Aug 2025 – present*

### Stereo Vision-Based ARAT Automation as Paralysis Rehabilitation Benchmark

*Tsinghua University, Hong Lab* · GROUP LEADER

- Designed the system architecture for scoring the Action Research Arm Test from stereo camera video, combining binocular stereo vision with CAD-based 6D pose estimation; implementation is ongoing
- Proposed tensor decomposition for dimensionality reduction in motor-imagery decoding
- Part of EBR, a minimally invasive BCI rehabilitation system funded as a key project by the Tsinghua Innovative Research Commission; supported ECoG data collection from patients

## COURSEWORK

GRADED Linear Algebra · Calculus · Probability & Statistics · Fundamentals of Computer Programming (Python) · Biostatistics · Data Science in Microbiology

## SKILLS

INTERPRETABILITY TransformerLens · activation patching · ablation and causal intervention

MACHINE LEARNING Python · PyTorch · LoRA fine-tuning of LLMs / VLMs · multi-model benchmarking

SIGNAL PROCESSING EEG & ECoG (FBCCA, TDCA, TRCA) · tensor decomposition · MATLAB / EEGLAB

LANGUAGES English (IELTS 8.0) · Mandarin (native) · Arabic (elementary)

## TALKS

*Jun 2026*

**Active Use of Latent Tree-Structured Sentence Representation in Humans and Large Language Models**
*Journal club · Gao Lab Meeting*

*Nov 2025*

**NetFormer: An Interpretable Model for Recovering Dynamical Connectivity in Neuronal Population Dynamics**
*Journal club · Hong Lab Meeting*

*Oct 2025*

**Dream Engineering: From Targeted Memory Reactivation to Visual Reconstruction**
*Symposium · Tsinghua Academy Elite Scholars Program*

*Aug 2025*

**From Psychoacoustic Category Boundaries to Cortical Feature-Tuned Ensembles: Edward Chang's Research on Speech Perception**
*Journal club · Hong Lab Meeting*

*Mar 2025*

**Boolean Network Models: Toward Interpretable and Robust Biomodeling**

*Symposium · Tsinghua Academy Elite Scholars Program · with Zezhao Wu*